

List Project Packages

Project Package	One-line Description	Workflow	Functions Included	Example Output	AI/DS/ML Application
Credit Risk Decision Support with Explainability	Build a prototype to support small-loan risk assessment: predict default probability, assign risk band, explain reasons, and log reviewer decisions.	Loan application → feature extraction → default prediction → risk banding → reason codes → approve/review/reject suggestion → decision audit log	Credit risk banding Credit explainability/reason codes Decision audit trail Adverse action reason generation Optional model performance/fairness monitoring	<code>default_probability: 0.68;</code> <code>risk_band: high;</code> <code>suggestion: manual_review;</code> reasons: high debt-to-income ratio, multiple late payments, short employment history	DS: EDA, segment analysis; ML: Logistic Regression / Random Forest / XGBoost; Explainability: SHAP / feature importance / reason codes; Governance-lite: decision log, model version
Open Banking Transaction & Cashflow Insight	Simulate Open Banking transaction data, then categorize transactions, detect unusual spending, identify recurring bills/income, and generate cashflow insights.	Simulated bank connection → transaction normalization → categorization → recurring bill/income detection → unusual spending/cashflow warning → monthly insight summary	Simulated account aggregation; transaction normalization; transaction categorization; unusual spending detection; monthly spending summary; optional cashflow forecasting / low balance warning	<code>category: utilities;</code> <code>recurring: true;</code> insight: electricity bill is 28% higher than 3-month average; alert: possible bill spike	ML/NLP: transaction categorization; DS: spending trend, recurring pattern, income/bill detection; Anomaly: unusual spending, bill spike; AI optional: grounded LLM summary

AI-Assisted eKYC Exception Review	Do not build full eKYC; simulate low-confidence eKYC cases and support reviewer decisions with risk scoring, mismatch detection, checklist generation, and audit summary.	Simulated eKYC engine output → risk banding → mismatch detection → low-confidence case routing → reviewer checklist → audit summary	KYC risk banding; reviewer checklist generator; audit summary generation; optional synthetic identity risk scoring; optional case similarity search; decision audit trail	risk_band: medium; suggestion: manual_review; reasons: name mismatch, low face match score, new device; checklist: retake ID image, verify spelling, repeat liveness capture	Rule/ML-lite: KYC risk banding, pass/review/reject classification; LLM: checklist, audit summary, structured explanation; DS: data quality/reviewer consistency optional; Governance: audit trail, PII/guardrail checks
Payment Operations Intelligence	Simulate payment events/webhooks to detect stuck payments, duplicate/out-of-order webhooks, status inconsistency, and classify failed payment reasons.	Payment created → provider event/webhook → internal status update → failed/stuck/duplicate detection → failure reason classification → retry/manual-review suggestion	Failed payment reason classifier; stuck payment detection; payment status inconsistency detector; webhook duplicate/out-of-order detector; retry recommendation; payment health dashboard	payment_health: stuck_or_failed; failure_reason: provider_timeout; suggestion: retry_later_with_same_idempotency_key; severity: medium	ML: failed payment reason classifier; Anomaly: stuck payment, retry storm, missing webhook; DS: provider/method success rate, latency, error spikes; AI optional: root-cause summary from event timeline

Merchant Settlement & Reconciliation Assistant	Simulate reconciliation between internal payment records and provider/bank settlement files to detect missing transactions, amount mismatch, fee/refund mismatch, and generate exception reports.	Internal payment records + provider settlement statement + refund/fee records → transaction matching → mismatch detection → exception classification → severity scoring → reconciliation report	Internal-vs-provider transaction matching; auto-match vs manual-review routing; exception classification; exception severity scoring; aging exception monitor; root-cause assistant; optional case similarity search	match_status: matched_with_fee_difference; exception_type: fee_deduction; severity: low; explanation: provider amount is lower by expected fee	DS: exact/fuzzy transaction matching, duplicate detection; ML-lite: match confidence scoring, exception classification; Anomaly: unusual mismatch rate, provider/merchant anomaly; AI optional: root-cause explanation and daily recon report
Promotion/Cashback Abuse & Multi-account Detection	Simulate voucher/cashback/referral campaigns in an e-wallet or super app and detect multi-account abuse, referral fraud, voucher farming, cashback loops, and promo-refund abuse.	Campaign launched → users redeem voucher/cashback → transaction/refund/referral events generated → graph/entity features → abuse cluster detection → review queue	Promotion/cashback abuse; referral fraud; voucher farming; cashback loop; multi-account abuse; promo-refund combined abuse	abuse_risk: high; cluster_id: CL_42; reasons: 12 accounts share same device, repeated cashback-refund loop, circular referral graph	Graph analytics: shared device/IP/referral graph; Anomaly: abnormal redemption/refund pattern; Clustering: suspicious user groups; ML-lite: abuse risk scoring

